

This fictional city brief was created for teaching RAG retrieval and evaluation.

Lakemont is a fictional lakeside city where short storms cause street flooding and send polluted runoff into Lake Aster. The green roof and stormwater program focuses on the Hillcrest, West Dock, and Museum Quarter neighborhoods. These areas have many flat roofs, parking lots, and old combined sewers.

The main target is to capture the first 25 millimeters of rainfall during common storms. City engineers call this the first-flush problem because the first part of a storm often carries oil, dust, and litter from streets into the lake. Capturing that water improves lake quality and reduces sewer overflow.

Lakemont offers grants for green roofs on apartment buildings, schools, and public offices. A green roof must include a waterproof layer, soil medium, drought-tolerant plants, and safe maintenance access. The city does not require every building to participate, but larger renovations must submit a stormwater retention plan.

Permeable pavement is planned for West Dock parking lanes and two public squares near the museum. Unlike normal asphalt, permeable pavement lets rain pass through gaps into a stone layer below. The program pairs permeable pavement with rain gardens so water has more than one place to slow down.

Schools receive cisterns that collect roof water for gardens and cleaning outdoor equipment. The education department will use the cisterns as climate lessons, but the infrastructure goal is practical: reduce pressure on sewers during fast storms. The program expects the school cisterns to be small but visible examples.

The city makes a clear distinction between stormwater adaptation and emergency flood response. Green roofs, rain gardens, and permeable pavement reduce runoff before streets flood. Sandbags and temporary road closures are response tools after flooding has already started. Lakemont wants more investment before the emergency phase.

The budget is 4.2 million city credits for the first two years, including 1.5 million for private building grants. Success is measured by retained stormwater volume, fewer sewer overflow warnings, and lower bacteria counts near the West Dock swimming area. The plan warns that maintenance is essential because clogged pavement stops absorbing water.